

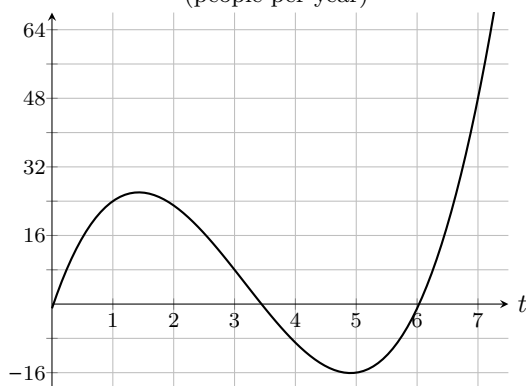
# The Definite Integral

## Today's Goals

- ◆ You should be able to recognize and use the notation  $\int_a^b f(x) dx$  for the definite integral.
- ◆ You should be able to think about the definite integral in three different ways: as a limit of left-hand or right-hand sums, as a signed area, and as a net change of a rate function.
- ◆ You should be able to use the graphical interpretation of the definite integral to evaluate  $\int_a^b f(x) dx$  when  $f$  is a function whose graph is simple or has useful symmetry.

1. The population of a tiny island country changes over time, as people immigrate and emigrate. Suppose  $f(t)$  gives the rate of change of the population, where  $t$  is measured in years.

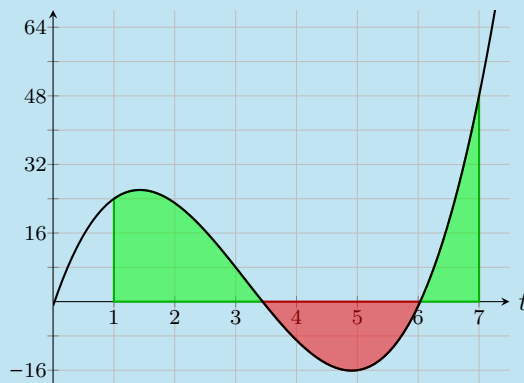
$f(t)$  = rate of change of island's population  
(people per year)



- (a) How would you visualize the exact net change in population between  $t = 1$  and  $t = 7$ ?

## Solution

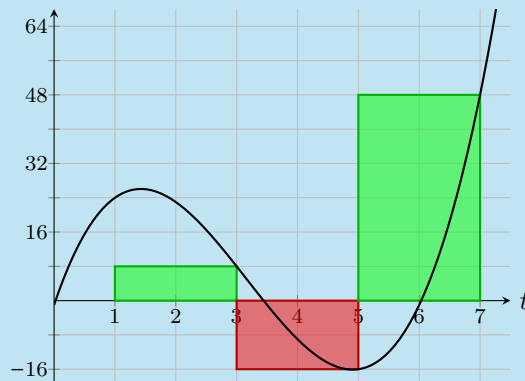
We can visualize the exact net change in population between  $t = 1$  and  $t = 7$  as the signed area between the curve and the  $x$ -axis. That's the area of the green regions below minus the area of the red region:



- (b) Use a right-hand sum with 3 rectangles to estimate the net change in population between  $t = 1$  and  $t = 7$ .

### Solution

Here is a picture of  $R_3$ :



From this, we can see  $R_3 = 8 \cdot 2 + (-16) \cdot 2 + 48 \cdot 2 = \boxed{80}$  (representing a net gain of 80 people).

- (c) Was the population larger at time  $t = 1$  or time  $t = 7$ ? How do you know?

### Solution

From our picture in (a), we can see that the net change from  $t = 1$  to  $t = 7$  is positive; therefore, the population was larger at  $\boxed{t = 7}$ .

- (d) Suppose the population at time  $t = 2$  was 2000 people. Write an expression involving a definite integral which gives the number of people at time  $t = 4$ .

### Solution

Definite integrals always tell us about net change, so let's first find the net change between  $t = 2$  (when we know the population) and  $t = 4$ : that's  $\int_2^4 f(t) dt$ . As an equation, we've just said

$$(\text{population at } t = 4) - (\text{population at } t = 2) = \int_2^4 f(t) dt,$$

so

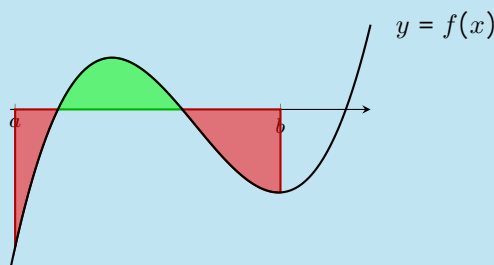
$$(\text{population at } t = 4) = (\text{population at } t = 2) + \int_2^4 f(t) dt$$

$$= \boxed{2000 + \int_2^4 f(t) dt}$$

## The Definite Integral

We have three major ways of thinking about the definite integral  $\int_a^b f(x) dx$ :

1. If  $a < b$ , we visualize it as the signed area under  $f$  from  $a$  to  $b$ .

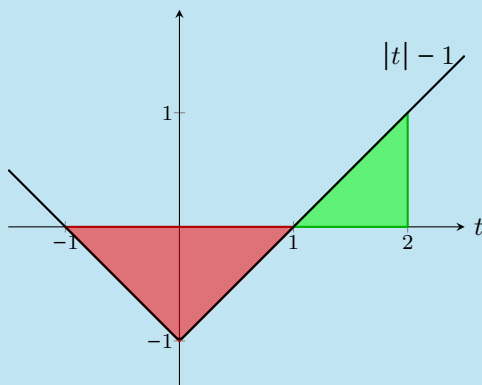


2. If  $f$  is a rate function, then we interpret  $\int_a^b f(x) dx$  as the net change in amount from  $x = a$  to  $x = b$ .
3.  $\int_a^b f(x) dx$  is actually defined to be  $\lim_{n \rightarrow \infty} R_n$  or  $\lim_{n \rightarrow \infty} L_n$ .

2. (a) Evaluate  $\int_{-1}^2 (|t| - 1) dt$ .

### Solution

Graphically, this is the signed area under  $y = |t| - 1$  from  $t = -1$  to  $t = 2$ , pictured below:



The red region is a triangle of width 2 and height 1, so its area is  $\frac{1}{2}(2)(1) = 1$ ; the green region is a triangle of width 1 and height 1, so its area is  $\frac{1}{2}(1)(1) = \frac{1}{2}$ . Therefore,  $\int_{-1}^2 (|t| - 1) dt = -1 + \frac{1}{2} = \boxed{-\frac{1}{2}}$ .

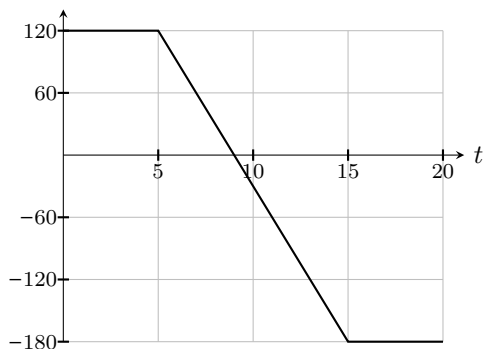
- (b) Suppose  $|t| - 1$  gives the rate (in gallons per hour) of water flowing into and out of a tank  $t$  hours after noon on a particular day. Which of the following is a correct interpretation of  $\int_{-1}^2 (|t| - 1) dt$ ?
  - A. It is the amount of water in the tank at 2 pm.

B. It is the amount of water in the tank at 11 am.

(C) It is (amount of water in the tank at 2 pm) – (amount of water in the tank at 11 am).

D. It is (amount of water in the tank at 11 am) – (amount of water in the tank at 2 pm).

3.



The function  $r(t)$  graphed to the left gives the rate at which water is flowing into or out of a backyard swimming pool. Time  $t$  is measured in hours, and the rate  $r(t)$  is measured in gallons per hour. At time  $t = 0$ , there are 700 gallons of water in the pool.

(a) When is the amount of water in the pool greatest? At that time, how much water is in the pool?

### Solution

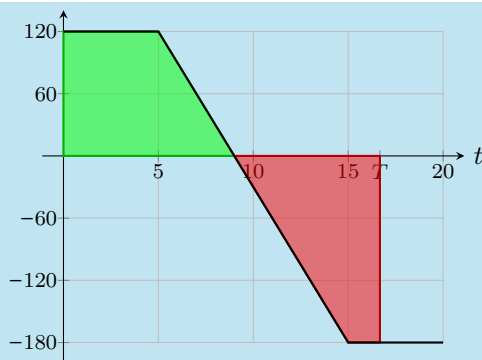
When  $r(t) > 0$ , the pool is gaining water; when  $r(t) < 0$ , the pool is losing water. So, the amount of water in the pool must be greatest at the time when  $r(t)$  switches from positive to negative; this is simply the  $t$ -intercept of the graph. To find this  $t$ -intercept, we can use the fact that the graph of  $r(t)$  is linear on  $[5, 15]$  with a slope of  $\frac{-180 - 120}{15 - 5} = -30$ . So, its equation is  $y - 120 = -30(t - 5)$ . The  $t$ -intercept is where  $y = 0$ , so where  $0 - 120 = -30(t - 5)$ , which happens at  $t = 9$ . Therefore, the amount of water in the pool must be greatest at  $t = 9$ . Moreover,

$$\begin{aligned} (\text{amount of water in pool at } t = 9) &= (\text{amount of water in pool at } t = 0) + \int_0^9 r(t) dt \\ &= 700 + (\text{area under graph of } r(t) \text{ from } t = 0 \text{ to } t = 9) \\ &= 700 + 840 \\ &= \boxed{1540 \text{ gallons}} \end{aligned}$$

(b) Recall that, at  $t = 0$ , the pool contains 700 gallons of water. Is there ever another time  $T$  at which the pool contains 700 gallons of water? If so, what is  $T$ ? If not, why not?

### Solution

As we saw in (a), the pool starts out with 700 gallons of water and has gained an additional 840 gallons by  $t = 9$ . After this, the pool starts losing water. We can see that, by  $t = 20$ , it has lost more water than it gained, so there must have been some time before  $t = 20$  when it had exactly 700 gallons. Graphically, we're looking for the time  $T$  when the signed area from  $t = 0$  to  $t = T$  is exactly 0:



That is, the area of the red region must be exactly the same as the area of the green region, which is 840. The portion of the red region from  $t = 9$  to  $t = 15$  is a triangle of width 6 and height 180, so it has area  $\frac{1}{2} \cdot 6 \cdot 180 = 540$ . Therefore, we want the portion of the red region from  $t = 15$  to  $t = T$  to have area  $840 - 540 = 300$ . This portion is a rectangle of height 180, so its width must be  $\frac{300}{180} = \frac{5}{3}$ . Therefore,  $T = 15 + \frac{5}{3} = \boxed{\frac{50}{3}}$ .

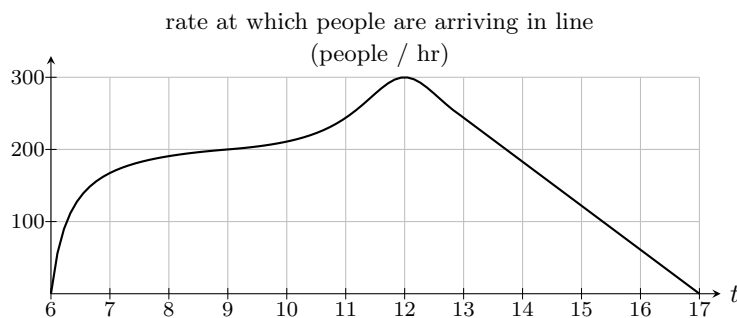
- (c) How much water is in the pool at time  $x$ ? (You may leave your answer in terms of a definite integral.)

### Solution

The amount of water in the pool at time  $x$  is (amount of water in the pool at  $t = 0$ ) + (net change in amount of water from  $t = 0$  to  $t = x$ ), or  $700 + \int_0^x r(t) dt$ .

4. In the wake of the Great Depression (around 1932), many Americans lost their jobs. At that time, unemployment insurance was not in place (that began in 1936). Many people stood in lines at soup kitchens, lines so long that they wrapped around city blocks.

The graph below is a graph of  $r(t)$ , the rate that people are arriving to get in line at a Detroit soup kitchen. Time  $t$  is given in hours since midnight, so  $t = 6$  is 6 am. (Nobody is in line at 6 am.) Soup is served between the hours of 9 am and 5 pm, and the rate at which people can be served is 200 per hour.



- (a) Approximately how long is the line when the soup kitchen opens?

### Solution

This is exactly the number of people who have arrived between 6 am and 9 am, which is

$\int_6^9 r(t) dt$ . If we just eyeball the graph, it looks like the area under  $r(t)$  from  $t = 6$  to  $t = 9$  is about 5 grid squares; each grid square represents 100 people, so there would be  $\approx 500$  people.

- (b) Approximately how long is the line at noon?

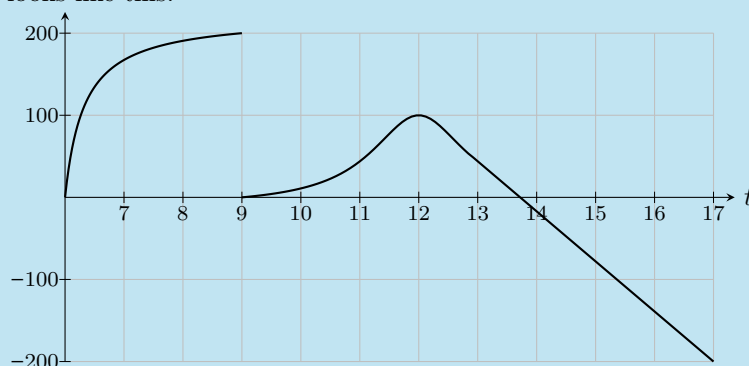
### Solution

We can think of this as (number of people who arrived between 6 am and noon) – (number of people who were served between 6 am and noon). The number of people who arrived between 6 am and noon is  $\int_6^{12} r(t) dt$ , which looks like about 12 grid squares, or 1200 people. People are only served between 9 am and 5 pm, so the number of people served by noon is  $(200 \text{ people / hour})(3 \text{ hours}) = 600$  people. Therefore, the length of the line at noon is about  $1200 - 600$ , or  $\approx 600$  people.

- (c) At approximately what time is the length of the line increasing most rapidly?

### Solution

From 6 am to 9 am, the rate at which the line increases is exactly the rate at which people arrive. But from 9 am until 5 pm, people are also being served at a rate of 200 people per hour, so the rate at which the line increases is  $r(t) - 200$ . Graphically, the rate at which the line increases looks like this:



So, the line is increasing most rapidly at (or right before) 9 am.

- (d) At approximately what time is the line the longest?

### Solution

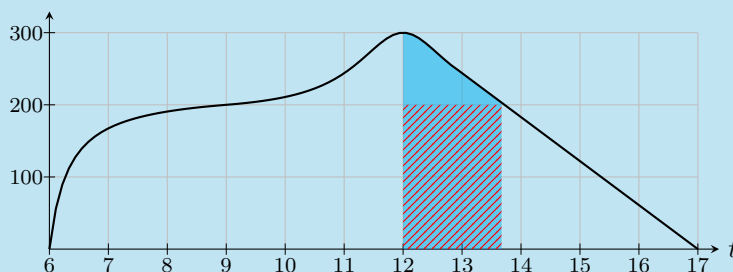
As long as the rate at which people are getting in line is higher than the rate at which people are served, the line continues growing. This is the case until about 1:40 pm; after that, the line begins to shrink because people arrive at a rate which is lower than the rate at which people are being served. So, the line is longest at about 1:40 pm.

- (e) When the line is the longest, approximately how many people are in line?

### Solution

This is asking for the length of the line around 1:40 pm, and we could approach this in exactly the same way as (b): it's (number of people who arrived between 6 am and 1:40 pm) – (number of people served between 6 am and 1:40 pm).

However, since we already estimated that there were 600 people in line at noon, it's a little more efficient to just estimate how the line changes between noon and 1:40 pm. This is (number of people who arrived between noon and 1:40 pm) – (number of people served between noon and 1:40 pm). We can picture the number of people who arrived between noon and 1:40 pm as the area under  $r(t)$  between noon and 1:40 pm (shaded in blue below), while the number of people served between noon and 1:40 pm can be pictured as the area under 200 between noon and 1:40 pm (shaded in red below). Therefore, the amount the line increased is the area just in blue, which is about 100 people. So, at 1:40 pm, there about  $600 + 100$ , or  $\approx 700$  people.



- (f) People at the end of the line routinely went home without food. How many people were left in line at closing time?

### Solution

Since we already estimated that there were about 700 people in line at 1:40 pm, let's figure out what happens after that. From 1:40 to 5 pm, the number of people who join the line is  $\int_{13+\frac{2}{3}}^{17} r(t) dt$ , which is about 330. The number of people who are served is (200 people per hour)  $(3 + \frac{1}{3} \text{ hours})$ , or about 670 people. So, the number of people in line at 5 pm is about  $700 + 330 - 670 \approx 360$ .

- (g) If there were enough food left to feed all people in line at 5 pm, at approximately what time would the work of the soup kitchen be done?

### Solution

Since there are about 360 people left in line and people can be served at 200 people per hour, it would take about  $\frac{360 \text{ people}}{200 \text{ people per hour}} = 1.8$  hours to feed them. 1.8 hours after 5 pm is  $\approx 6:48 \text{ pm}$ .

Express the following quantities in terms of definite integrals.

- (h) The length of the line at 8 am.

### Solution

The length of the line at 8 am is just the number of people who arrive between 6 am and 8 am (since the soup kitchen has not started serving anyone yet), or  $\int_6^8 r(t) dt$ .

- (i) The length of the line at 10 am.

### Solution

The length of the line at 10 am is (number of people who arrive between 6 am and 10 am) – (number of people served by 10 am), or  $\int_6^{10} r(t) dt - 200$ .

- (j) The amount of time a person arriving at 10 am must wait to be served.

### Solution

If a person arrives at 10 am, there are  $\int_6^{10} r(t) dt - 200$  people already in line (by (i)). Using the same reasoning as in (g), the amount of time the person must wait is  $\frac{\int_6^{10} r(t) dt - 200}{200}$  hours.